

WCS417 drought rescue across 247 *Arabidopsis* accessions

Reanalysis of the natural-variation screen, including the replicate runs

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Unpublished data of the collaborating laboratory — not for distribution

The question before the GWAS

- The screen is a factorial for every accession:
 - mock / WCS417 × drought / non-stress × shoot / root.
- A GWAS needs **one number** per accession.
- Six readings were mapped separately:
 - biomass, gain, % increase, % rescue, % loss, rescuer label.
- Each reading gave different candidate genes, and no accession lost rescue.

Two questions

1. Which of these numbers *is* the rescue of an accession?
2. Does it replicate from one run to the next?

Part I answers question 1 from the first workbook. Part II answers question 2 from the full per-plant workbook.

The data

First workbook (GWAS input)

- Per-plant weights for WCS417 + drought only (1,841 plants).
- The other cells appear only as averages inside the derived traits.
- easyGWAS output: 28 candidate genes, 40 rows.

Second workbook (NS+DS.xlsx)

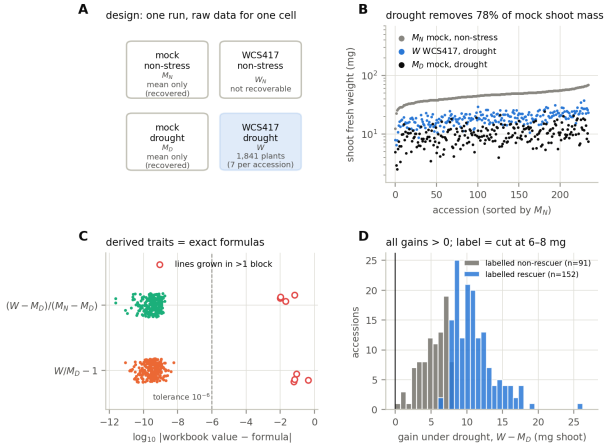
- 8,820 plants: all four cells, 247 accessions.
- Drought run 1: all accessions.
- Drought runs 2 and 3: 50 accessions,
 - 43 of them labelled *non-rescuers*.
- Col-0 grown in 7-plant blocks in every cell and run.

Notation: W = WCS417 drought, M_D = mock drought, M_N = mock non-stress, W_N = WCS417 non-stress (all cell means).

Part I

Which number is the rescue of an accession?

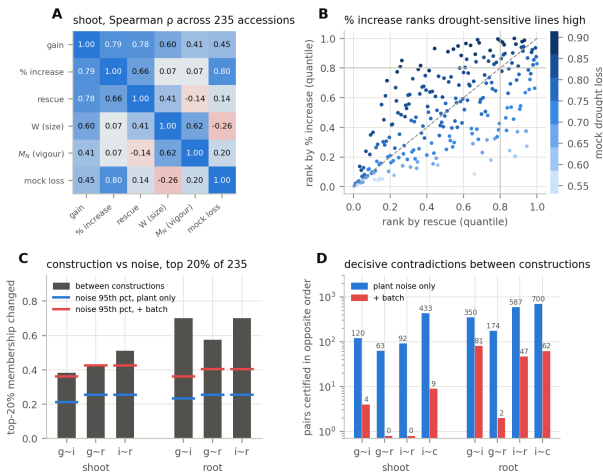
Every derived trait is an exact formula in three cell means



- The formulas fit all 235 complete accessions to 10^{-10} :
 - rescue

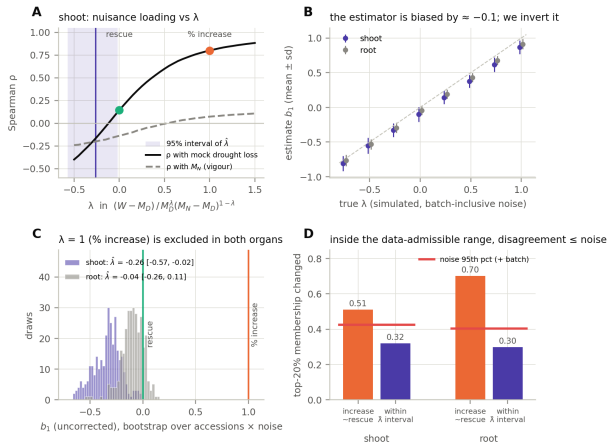
$$= (W - M_D)/(M_N - M_D)$$
 - increase = $W/M_D - 1$
- The second workbook **confirms** the reconstructed mock means by measurement.
- Three corrections:
 - % loss uses **no inoculated plants**.
 - Table 2 “tolerance” is actually loss.
 - “Non-rescuer” is a cut at 6–8 mg; every accession gains.

The readings disagree about who rescues best



- Top-20% overlap between readings:
 - Jaccard 0.32–0.45 in shoot;
 - 0.18–0.27 in root.
- Each reading carries a different nuisance:
 - % increase ranks **drought sensitivity** ($\rho = 0.80$);
 - gain tracks plant size ($\rho = 0.60$);
 - rescue is the least loaded, but the noisiest.
- Under plant noise, two readings certify up to 700 pairs of accessions in **opposite order**.

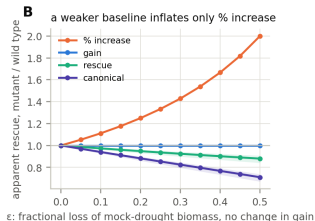
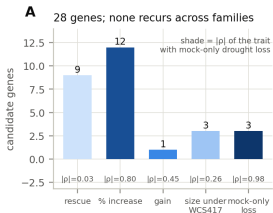
Letting the data choose the reading



$$f_{\lambda} = \frac{W - M_D}{M_D^{\lambda} (M_N - M_D)^{1-\lambda}}$$

- $\lambda = 1$ is % increase; $\lambda = 0$ is rescue.
- Choosing λ means choosing a mechanism: does WCS417 multiply the stressed biomass, or restore a share of what drought removed?
- Estimated from the data: shoot $\hat{\lambda} \approx -0.15$, root $\hat{\lambda} \approx 0.05$. **% increase is excluded**; rescue lies at or inside the interval.

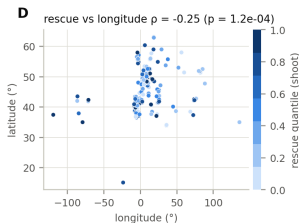
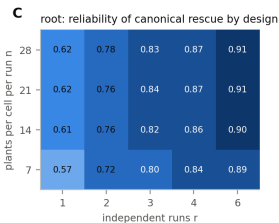
Candidate genes, sorted by what they are loci *for*



- 28 genes; **none** recurs across trait families.

- By the trait family each was found in:

- 3 from mock-only loss: MORN4, TPS21, AT2G07213;
- 12 from % increase: at risk of being drought-sensitivity loci;
- 9 from rescue.

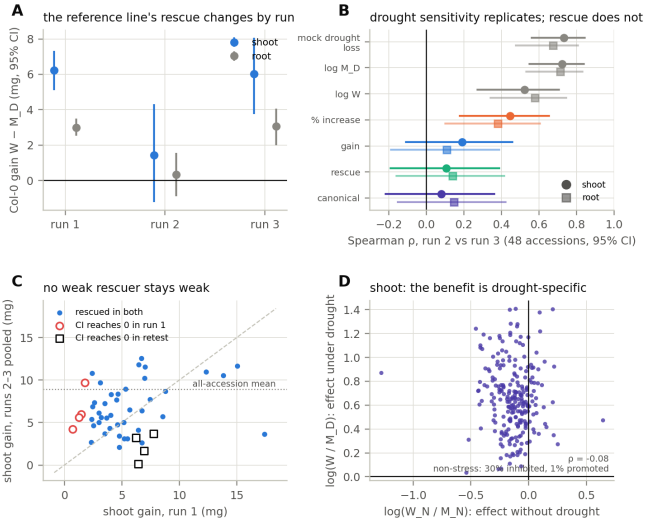


- Mutant check (panel B): a mutant that is only weaker under drought looks up to 2× “better rescued” under % increase, and unchanged under gain.

Part II

The full per-plant workbook: three runs and a non-stress control

The replicate runs at a glance



The reference line is not stable across runs

Col-0, shoot	run 1	run 2	run 3
gain (mg)	6.2	1.4	6.0
95% CI	[5.1, 7.3]	[-1.2, 4.3]	[3.8, 8.1]

Root: 3.0, 0.3, 3.1 mg.

- Same genotype, same protocol.
- In run 2, Col-0 would have been called a **non-rescuer**.
- Within a run, batch is largest in mock-drought plants (block CV 0.18).
- Without drought, mock and inoculated trays moved together ($r = 0.66$ – 0.88), so batch cancelled. Under drought they did not ($r \approx 0$).

What survives an independent rerun?

Spearman ρ between run 2 and run 3 for the same 48 accessions. No modelling is involved, and neither run was used to select them.

	shoot ρ [95% CI]	root ρ [95% CI]
mock drought loss (no bacterium)	0.73 [0.56, 0.84]	0.68 [0.48, 0.81]
log M_D	0.72 [0.55, 0.84]	0.71 [0.53, 0.83]
log W	0.52 [0.27, 0.71]	0.58 [0.34, 0.74]
% increase	0.45 [0.18, 0.65]	0.38 [0.10, 0.61]
gain	0.19 [-0.11, 0.46]	0.11 [-0.19, 0.39]
rescue	0.11 [-0.19, 0.39]	0.14 [-0.16, 0.42]

- **Drought sensitivity replicates; the benefit from WCS417 does not.**
- % increase does moderately well only because it ranks drought sensitivity.

No weak rescuer stays weak

- 50 apparent weak rescuers were regrown twice.
- Gain compatible with zero:
 - 5 accessions in run 1;
 - 4 *different* accessions in the pooled retest;
 - **0 in both**, and no negative retest estimate.
- Per-1, the best loss-of-rescue candidate, was rescued in both retests: 5.3 and 3.2 mg, both intervals above zero.
- The regrown accessions moved 26% (shoot) and 32% (root) of the way back to the population mean. This is regression to the mean after selecting on one noisy run.

Conclusion

There is no confirmed loss-of-rescue accession. This is consistent with the **redundancy** hypothesis: WCS417 helps essentially every genotype.

The benefit is specific to drought

Without drought

- WCS417 slightly **inhibits** shoot growth: mean -12% .
- 30% of accessions are significantly inhibited; 1% promoted.
- Root is unchanged on average.

Under drought

- 95% (shoot) and 99% (root) of accessions are promoted.
- The effect is **uncorrelated** with the non-stress effect ($\rho = -0.08, -0.01$).

- This is not general growth promotion carried into drought. It differs from plate assays without stress (Wintermans et al. 2016), where nearly all accessions were promoted.
- The inoculation $\times$ water interaction is not a clean phenotype either: it correlates 0.79–0.82 with drought loss.

How finely can the screen rank accessions?

Canonical rescue, shoot	median certifiable margin	tiers resolved
measured plant noise	2.0 SD	8
+ within-run batch (Col-0 blocks)	3.3 SD	3
+ run-to-run noise (runs 2 vs 3)	4.6 SD	2

- “Certifiable margin”: the smallest difference at which two accessions could be declared equivalent, in units of the between-accession spread.
- The screen can say that accessions differ. It cannot say that they rescue equally.
- Once realistic run-to-run noise is included, a single run separates **2–3 levels** of rescue, not 235 ranks. Root resolves 3.

What this means for the GWAS

- 1. The current candidate genes are hits on a single-run phenotype.**
 - Rescue has a run-to-run reliability of about 0.1–0.2.
 - Expect poor replication, except for drought-sensitivity loci.
- 2. Drought sensitivity is the solid result.**
 - Mock drought loss replicates at $\rho \approx 0.7$.
 - A drought-sensitivity GWAS is publishable now; the phenotype table is prepared.
- 3. A rescue GWAS needs a phenotype averaged over several runs.**
 - Spearman–Brown: reliability 0.6 needs about 6 runs (at 0.2) to 14 runs (at 0.1).
- 4. The biology survives and sharpens.**
 - The benefit is drought-specific and reliably positive.
 - Accessions do not reproducibly differ in how much they are helped.

Design for the next screen

- **Independent runs, not more plants.**
 - A second run is worth more than four times the plants in one run.
- **Co-locate mock and inoculated plants** of each accession on the same tray.
 - This already happened without drought, and the batch cancelled.
- **Control drought severity between runs.**
 - Col-0 mock-drought biomass varied 16–22 mg between runs.
- **Keep Col-0 in every block**, record block membership, and model block.
- **Fix the phenotype before mapping:** loss-normalised rescue, averaged over runs, with kinship and (in root) $\log M_N$ as covariates.
- **Grow mutants in the same run as their wild type**, and report gain and rescue alongside % increase.

Open questions for the lab

1. Were runs 2 and 3 chosen to confirm the apparent non-rescuers?
2. Under drought, were mock and inoculated plants on separate trays?
3. How was drought severity set and monitored in each run?
4. Was the non-stress experiment a single run, and grown at the same time as run 1?
5. Were the iron-uptake mutants grown alongside their wild type, and which metric showed “higher rescue”?
6. Can we get the genotypes (1001 Genomes IDs are already in the sheet), so the drought-sensitivity GWAS can be rerun with kinship?

Caveats

- The replicated accessions were selected as weak rescuers. Their run-1 gains still spread 91% as widely as the full panel's.
- Two runs on 48 accessions give wide intervals; the tables show them.
- Non-stress was grown once, so its run-to-run reproducibility is unknown.
- No genotypes were available: no association was rerun or tested.
- The λ family is a model. In root, the data reject its exact scale invariance.
- Bootstrap-based quantities move slightly between reruns. Accessions near a threshold (e.g. Choto-1, WI-0) can flip.

Code: `validation/experiments.py` (E1–E12) and `experiments_full.py` (F1–F7), seed 417; every number is read from `results/*.json`.